

THE EUROPEAN CENTRAL BANK

Unified Mentor — Applied Analytics Program

Customer Engagement & Product Utilization Analytics for Retention Strategy

A Behavioral and Relationship-Strength Approach to Customer Churn

Research Paper

Dataset: European Bank Customer Records (N = 10,000)

Prepared for: Government & Executive Stakeholders

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Abstract

This research paper examines customer retention at a major European retail bank through the lens of behavioral engagement and product utilization rather than static demographic or financial attributes. Analyzing 10,000 anonymized customer records across France, Germany, and Spain, the study finds that engagement depth — measured through account activity status and the breadth of products held — is a substantially stronger predictor of churn than balance or income alone. Customers holding a single product churn at more than twice the rate of multi-product customers (27.7% vs 12.8%), and inactive members churn at nearly double the rate of active members (26.9% vs 14.3%). Critically, the analysis identifies a segment of financially attractive but behaviorally disengaged customers — high-balance, inactive account holders — who churn at 30.5%, the highest rate of any segment analyzed except customers holding three or more products, a small group associated with likely dissatisfaction or forced cross-sell. A composite Relationship Strength Index, combining activity status and product depth, shows a near-monotonic reduction in churn from 34.5% at the weakest tier to 9.1% at the strongest, confirming that relationship depth — not wealth — is the dominant lever available to retention teams. The paper concludes with a set of KPIs, a segmentation framework, and concrete recommendations for engagement-driven retention strategy, product bundling, and early-warning detection of silent churn among premium customers.

1. Introduction and Background

Retail banks have traditionally approached customer retention through a demographic and financial lens — segmenting customers by age, income, or account balance and assuming that wealthier or older customers are inherently more loyal. This assumption is increasingly unreliable. A customer may present a strong financial profile on paper — a high balance, a stable salary, a long tenure — while being functionally disengaged from the bank's products and services. Such customers are the source of what this paper terms silent churn: attrition that is not preceded by any visible signal in traditional financial metrics, because the warning signs live in behavioral data rather than balance-sheet data.

This project reframes the retention problem around two behavioral dimensions: engagement (whether a customer is an active member of the bank's ecosystem) and product utilization (how many distinct products a customer holds, which acts as a proxy for relationship depth and switching cost). By evaluating churn through this lens, the analysis aims to give the bank's retention, marketing, and product teams a more actionable and earlier-warning view of risk than demographics alone can provide.

2. Problem Statement

Despite collecting granular data on customer engagement and product usage, banks frequently lack the quantitative tools to translate this data into action. Three gaps motivate this study:

- Limited quantitative insight into which specific behaviors — activity status, product count, product mix — are most strongly associated with churn.
- Ambiguity over whether product depth genuinely reduces churn, or whether the relationship is confounded by other factors such as income or geography.
- No systematic evidence on whether high account balances alone are sufficient to guarantee customer loyalty.

In the absence of this evidence, retention strategies tend to default to generic, balance-driven interventions (e.g., relationship-manager outreach reserved for the wealthiest clients) that may miss large populations of at-risk customers who do not appear financially distressed.

3. Project Objectives

3.1 Primary Objectives

- Evaluate the quantitative relationship between customer engagement (activity status) and churn.
- Measure the retention impact of product count and product mix (single- vs multi-product holdings).
- Identify disengaged yet high-value (high-balance) customers who represent a silent churn risk.

3.2 Secondary Objectives

- Support the design of engagement-driven retention strategies.
- Improve product bundling and cross-sell decisions using empirical churn patterns.
- Reduce silent churn among premium customers through earlier behavioral detection.

4. Dataset Description

The dataset comprises 10,000 customer records from a European retail bank operating in France, Germany, and Spain, with no missing values across any field. Each record captures demographic attributes, account and engagement indicators, and a binary churn label (Exited).

Column	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer creditworthiness score
Geography	Country of residence — France, Spain, or Germany
Gender	Male / Female
Age	Customer age in years
Tenure	Years as a bank customer
Balance	Account balance (EUR)

Column	Description
NumOfProducts	Number of bank products held
HasCrCard	Credit card ownership (1 = yes, 0 = no)
IsActiveMember	Activity indicator (1 = active, 0 = inactive)
EstimatedSalary	Estimated annual salary (EUR)
Exited	Churn indicator — target variable (1 = churned)

4.1 Sample Composition

- Total customers analyzed: 10,000 — 0 missing values across all 14 fields.
- Overall churn rate: 20.4% (2,037 churned customers; 7,963 retained).
- Geographic split: France 50.1% (5,014), Germany 25.1% (2,509), Spain 24.8% (2,477).
- Average churned-customer balance (€91,109) is notably higher than retained-customer balance (€72,745) — an early signal that balance alone does not protect against churn.

Overall Customer Churn Distribution (N = 10,000)

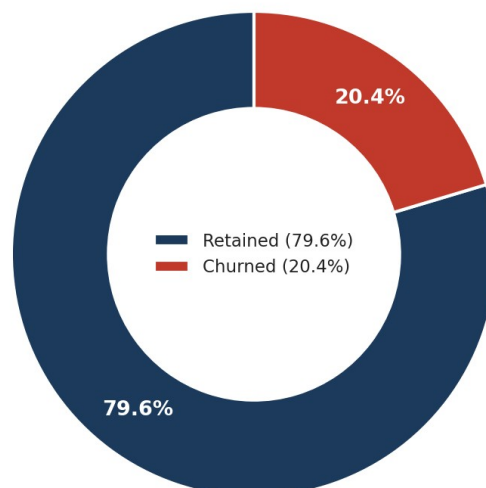


Figure 1. Overall churn distribution across the 10,000-customer sample.

5. Analytical Methodology

5.1 Data Ingestion & Validation

- Loaded and validated the full dataset; confirmed zero missing values.
- Verified binary fields (HasCrCard, IsActiveMember, Exited) contain only 0/1 values.
- Confirmed churn labeling consistency and class balance (79.6% retained / 20.4% churned).

5.2 Engagement Classification

Each customer was assigned to one of four behavioral engagement profiles, combining activity status, product depth, and balance level:

- Active Engaged — active member holding 2 or more products (n = 2,588).
- Active Low-Product — active member holding a single product (n = 2,563).
- Inactive Disengaged — inactive member with balance below the top quartile (n = 2,493).
- Inactive High-Balance — inactive member with balance in the top quartile ($\geq$ €127,644) (n = 2,356).

5.3 Product Utilization Analysis

- Churn rate computed for each product count (1, 2, 3, 4 products).
- Single-product holders compared against multi-product holders.
- Product depth cross-tabulated against churn to isolate the relationship-depth effect.

5.4 Financial Commitment vs Engagement Analysis

- Customers split into high-balance (top quartile, $\geq$ €127,644) vs standard-balance segments.
- Balance status cross-analyzed with activity status to form a 2×2 risk quadrant.
- At-risk premium customers identified as high-balance and inactive.

5.5 Retention Strength Assessment

A composite Relationship Strength Index (RSI) was constructed by summing weighted points for activity status (+2 if active), product depth (+2 if 2 or more products), and credit card ownership (+1 if held), producing a 0–5 scale. Churn stability was then measured across each RSI tier.

6. Findings

6.1 Engagement and Churn

Activity status alone produces a clear and substantial split in churn behavior. Inactive members churn at 26.9%, nearly double the 14.3% rate observed among active members. When engagement is combined with product depth into the four behavioral profiles defined above, the gradient becomes even sharper.

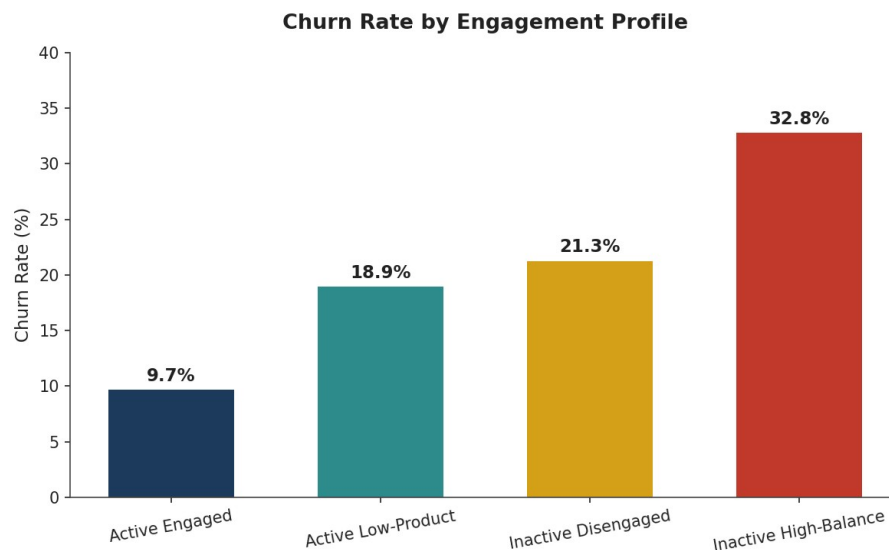


Figure 2. Churn rate by engagement profile, ranging from 9.7% (Active Engaged) to 32.8% (Inactive High-Balance).

Active Engaged customers — active members holding two or more products — churn at just 9.7%, the lowest of any segment. At the opposite extreme, Inactive High-Balance customers churn at 32.8%, more than three times the Active Engaged rate, despite (or arguably because of) their financial attractiveness. This is the clearest evidence in the dataset that engagement, not wealth, drives retention.

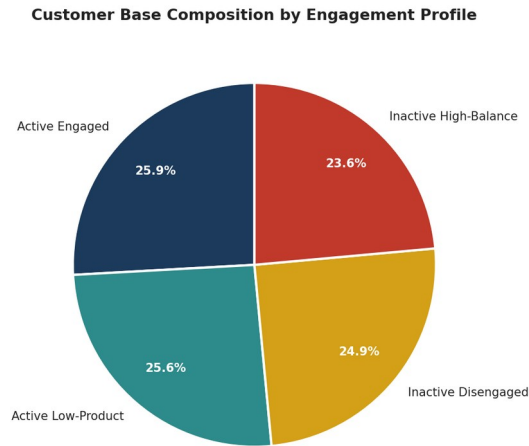


Figure 3. Customer base composition across the four engagement profiles — the base is nearly evenly split, meaning each segment represents a substantial retention opportunity.

6.2 Product Utilization and Churn

Product count shows the strongest and most policy-relevant relationship with churn in the dataset. Customers with a single product churn at 27.7%; churn drops to 7.6% for customers holding two products — the lowest rate observed for any product tier. Beyond two products, however, churn rises sharply: 82.7% for three products and 100% for four products, though these groups are small (266 and 60 customers respectively) and almost certainly reflect a different mechanism — likely customers pushed into additional products through aggressive cross-selling, or products bundled with accounts already earmarked for closure, rather than genuine multi-product loyalty.

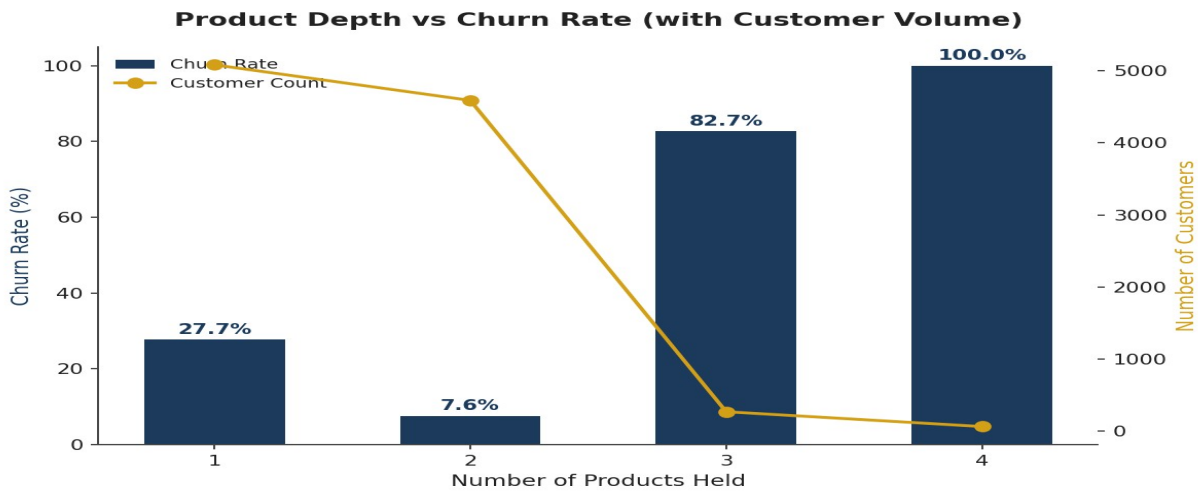


Figure 4. Churn rate by number of products held, overlaid with customer volume per tier.

Collapsing the data into single- vs multi-product holders confirms the core finding at a policy-relevant level of granularity: multi-product customers churn at 12.8% versus 27.7% for single-product customers — a more than two-fold difference.

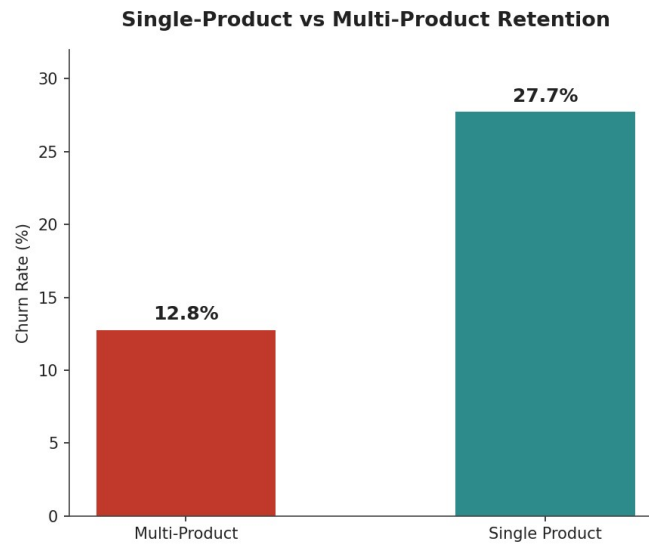


Figure 5. Single-product customers churn at more than double the rate of multi-product customers.

Credit card ownership, by contrast, shows almost no relationship with churn (20.2% for cardholders vs 20.8% for non-cardholders) — indicating that card issuance alone does not deepen the relationship enough to influence loyalty in this dataset.

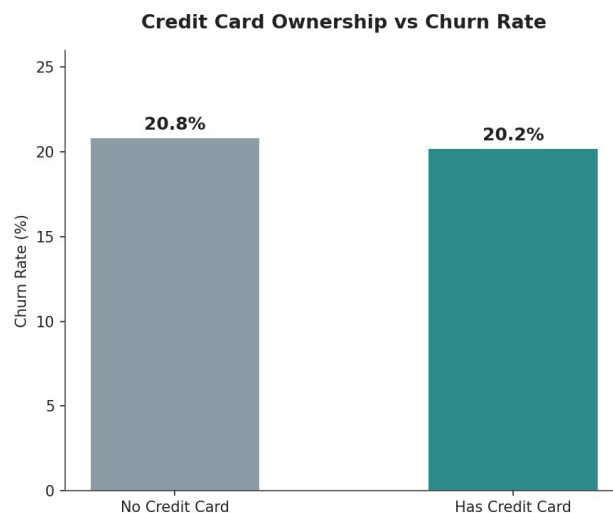


Figure 6. Credit card ownership is not a meaningful retention lever on its own.

6.3 Financial Commitment vs Engagement — Detecting Premium At-Risk Customers

Segmenting customers by both balance level and activity status reveals that engagement, not balance, is the dominant driver of risk within every balance tier. Among high-balance customers (top quartile), inactive members churn at 30.5% versus 16.9% for active members — a gap nearly as large as the gap observed among standard-balance customers (25.6% inactive vs 13.4% active). This demonstrates that a large balance does not offset the risk created by disengagement; if anything, high-balance inactive customers are the single riskiest segment of meaningful size in the dataset.

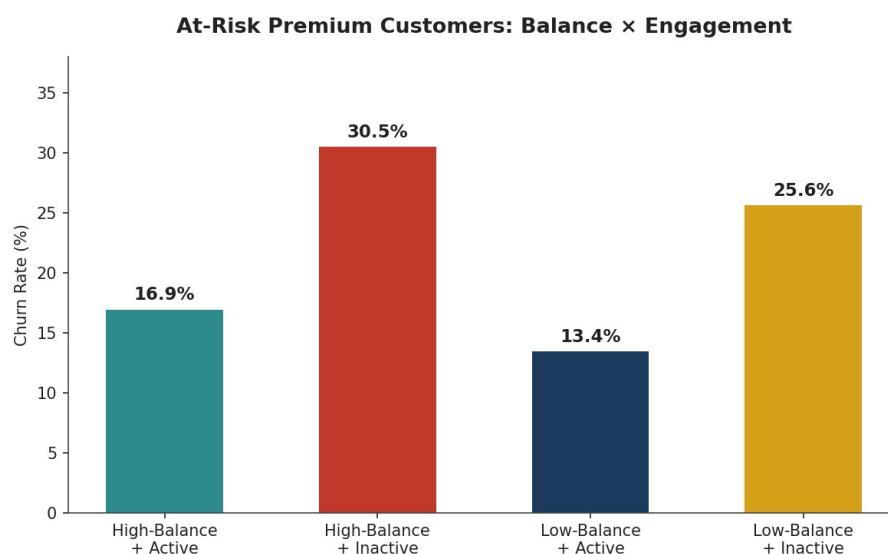


Figure 7. Churn rate across the balance × engagement risk quadrant. High-balance, inactive customers ($n = 1,247$) represent the clearest silent-churn population.

This 1,247-customer segment — high balance, inactive — is the practical definition of “silent churn risk” referenced in the problem statement: customers who look financially healthy in any standard portfolio review, yet are already behaviorally disengaged and churning at nearly one in three.

6.4 Retention Strength Assessment

The composite Relationship Strength Index (RSI), which combines activity status, product depth, and card ownership into a single 0–5 score, shows a near-monotonic decline in churn as relationship strength increases — from 34.5% at RSI 0 (weakest relationships) to 9.1% at RSI 5 (strongest). The one exception is a slight increase from RSI 0 to RSI 1 (34.5% → 37.5%), suggesting a small pocket of customers who hold a credit card in isolation without any other engagement signal, and who churn at the highest rate in the entire dataset.

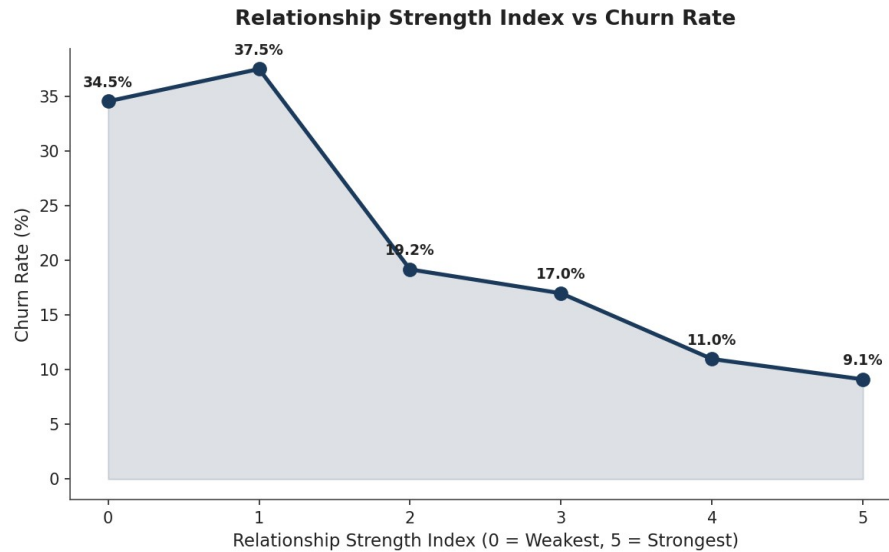


Figure 8. Churn rate declines sharply as the Relationship Strength Index rises, confirming that combined behavioral signals outperform any single metric.

6.5 Demographic and Geographic Context

While the project's primary lens is behavioral, demographic and geographic patterns provide useful context for targeting. Germany shows a markedly higher churn rate (32.4%) than France (16.2%) or Spain (16.7%), despite comparable product and engagement distributions, suggesting market-specific competitive or service factors are at play and warrant local investigation. Female customers churn at a higher rate than male customers (25.1% vs 16.5%). Churn also rises sharply with age through the 41–60 bracket (peaking at 56.2% for ages 51–60) before falling among customers over 60, a pattern consistent with life-stage-driven product switching (e.g., mortgage payoff, retirement account consolidation) rather than dissatisfaction alone.

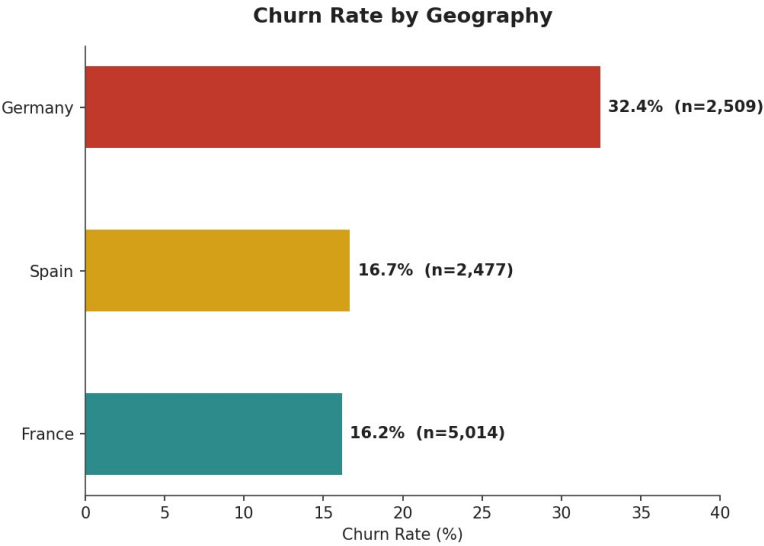


Figure 9. Germany's churn rate is roughly double that of France and Spain.

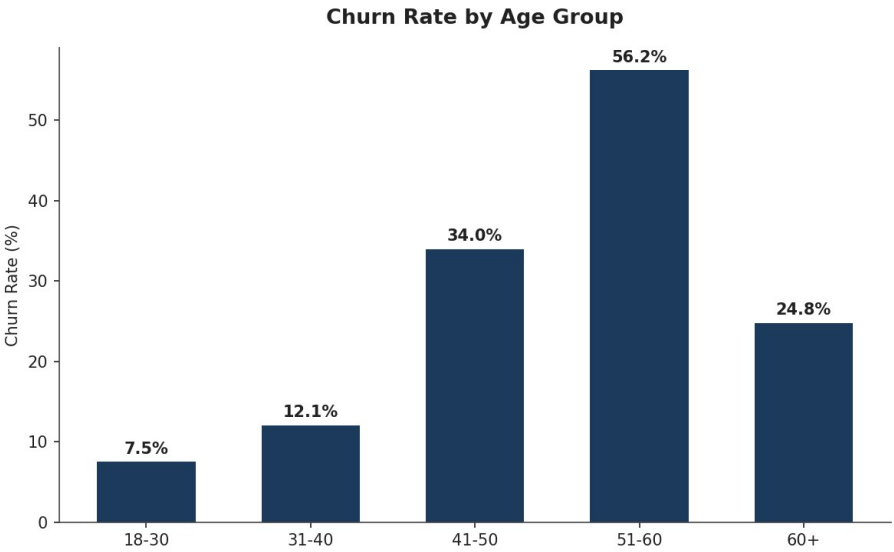


Figure 10. Churn rises through middle age and peaks in the 51–60 bracket before declining.

7. Key Performance Indicators (KPIs)

The following KPIs are proposed to operationalize the findings above into an ongoing retention monitoring framework, with baseline values computed from the current dataset.

KPI Name	Description	Baseline Value
Engagement Retention Ratio	Ratio of inactive-member churn to active-member churn	1.88× (26.9% vs 14.3%)
Product Depth Index	Churn rate for single- vs multi-product customers	27.7% vs 12.8%
High-Balance Disengagement Rate	Churn rate among top-quartile-balance, inactive customers	30.5% (n = 1,247)
Credit Card Stickiness Score	Churn-rate difference between cardholders and non-holders	0.6 pts (20.2% vs 20.8%) — negligible
Relationship Strength Index	Composite 0–5 score combining activity, product depth, card ownership, mapped to churn	34.5% (RSI 0) → 9.1% (RSI 5)

8. Key Insights Summary

- Engagement beats wealth: Activity status and product depth are stronger churn predictors than balance or salary; churned customers actually hold a higher average balance (€91,109) than retained customers (€72,745).
- The two-product sweet spot: Churn is minimized at exactly two products (7.6%); a third or fourth product is associated with sharply elevated churn, likely reflecting forced cross-sell rather than loyalty.
- Silent churn is real and measurable: 1,247 high-balance, inactive customers churn at 30.5% — a segment invisible to balance-only monitoring.
- Combined signals outperform single metrics: The Relationship Strength Index shows a clean, near-monotonic gradient from 34.5% to 9.1% churn, more actionable than any single variable alone.
- Credit cards alone don't build loyalty: Card ownership has almost no independent effect on retention (0.6-point difference).

- Germany is a geographic outlier: Its 32.4% churn rate is roughly double that of France and Spain and merits market-specific investigation.

9. Recommendations

9.1 Engagement-Driven Retention Strategy

- Deploy reactivation campaigns targeted specifically at inactive members, prioritized by balance tier — starting with the 1,247 high-balance, inactive customers identified as the top silent-churn segment.
- Build an early-warning score using activity-status transitions (active → inactive) as a leading indicator, rather than waiting for balance withdrawal or account closure signals.

9.2 Product Bundling Decisions

- Prioritize cross-sell toward a genuine second product (the empirically strongest retention tier) rather than pursuing indiscriminate product-count growth.
- Investigate the three- and four-product segments qualitatively — likely candidates include forced bundling at account opening or a mismatch between product terms and customer needs — before using these tiers as a cross-sell target.

9.3 Reducing Silent Churn Among Premium Customers

- Introduce relationship-manager check-ins triggered by the high-balance + inactive combination specifically, rather than balance alone.
- Extend the Relationship Strength Index into a live dashboard score so relationship teams can monitor tier migration (e.g., a customer moving from RSI 3 to RSI 1) as an actionable alert.

9.4 Market-Specific Action

- Commission a focused root-cause review of the German market's elevated churn rate, covering competitive positioning, service quality, and pricing relative to France and Spain.

10. Limitations and Future Work

- The dataset is cross-sectional (a single snapshot) rather than longitudinal; it cannot directly observe the sequence in which disengagement precedes churn, only their association.
- The three- and four-product segments are small (266 and 60 customers) and their extreme churn rates should be treated as directional rather than statistically robust without further investigation.
- The Relationship Strength Index weights (activity $\times 2$, products $\times 2$, card $\times 1$) were set heuristically for interpretability; a follow-on study could fit a regression- or tree-based model to derive statistically optimal weights and quantify predictor importance formally.
- Geographic differences (notably Germany) are documented but not explained by the available fields; qualitative or market-research follow-up is recommended.

11. Conclusion

This analysis reframes customer churn at the bank from a demographic and financial question into a behavioral and relationship-strength question. The evidence is consistent and substantial: engagement and product depth predict churn far more reliably than balance, salary, or card ownership. A composite Relationship Strength Index captures this relationship cleanly, moving from 34.5% churn at its weakest tier to 9.1% at its strongest — a range far wider than any single demographic cut in the dataset produces. Most importantly, the analysis surfaces a specific, addressable population of 1,247 high-balance, inactive customers who represent the clearest and most valuable silent-churn opportunity for the bank's retention teams. Shifting retention strategy from generic, balance-triggered outreach to engagement-driven, relationship-strength-triggered outreach represents the single highest-leverage change indicated by this data.